

AccessWatch

Multimodal Fusion for Forecasting of Humanitarian Access Risk in Ukraine

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Overview

1. Why Our Work Matters

2. Proposed Solution

3. Data and Experimental Setup

4. Findings

5. Limitations and Future Work

6. Demo

Why Our Work Matters



- Humanitarian access conditions can worsen rapidly in active conflict zones.
- A short-horizon warning system can help shift decision-making from reactive reporting toward proactive monitoring and prioritization.

Proposed Solution



Our solution: Forecast the state of a Ukrainian raion at future horizon $h \in \{1,2,3,4,5\}$ weeks ahead.

Predict:

- Any-event occurrence
- High-intensity conflict
- Event type forecasting

Datasets

- ACLED conflict-event history: structured georeferenced event records
- NASA FIRMS thermal anomaly data: satellite-derived point detections of thermal hotspots
- VIIRS Black Marble nighttime-light data: satellite-derived raster data capturing weekly illumination level, change, and observation quality, used as a proxy for disruption or recovery.
- Ukraine admin-2 boundaries: vector polygon boundary data
- Population, settlement, and infrastructure context layers: vector/raster geospatial context data
- Pre-2024 UNOSAT damage/disruption products: vector geospatial damage-assessment layers

Year: 2023-2024

High-intensity label rule:

High-intensity = 1 if future-week event count ≥ 10 OR fatalities ≥ 5 OR violence against civilians > 0 .

Event:

- Battle
- Violence Against Civilians
- Explosions/Remote violence
- Strategic Development
- Air Drone Strike

Data and Experimental Setup



How we evaluated the system

- Forecasting panel built over 139 raions and 104 weekly anchors
- Strict chronological split:
 - 78 training weeks
 - 13 validation weeks
 - 13 test weeks
- Two feature sets:
 - Base multimodal input
 - Lagged-ACLED-enhanced input
- Five model families:
 - Linear baseline
 - LightGBM
 - CatBoost
 - GRU
 - TCN
- Five missing-modality settings
- Total: 250 held-out test configurations

Why this matters:

This is not just a single-model demo; it is a structured benchmarking setup.

Findings



Table 1: Effect of adding lagged ACLED conflict-history variables under full input. Positive deltas indicate higher F1 and lower MAE.

Feature set	Any-event F1	High-intensity F1	Event-type macro-F1
Base multimodal input	0.7947	0.7748	0.6537
Lagged-ACLED-enhanced input	0.8194	0.8448	0.6963
Δ (Lagged – Base)	+0.0247	+0.0701	+0.0426

Table 2: Full-input classification performance averaged over forecast horizons. Best value in each column is bolded.

Model family	Base multimodal input			Lagged-ACLED-enhanced input		
	Any-event F1	High-intensity F1	Event-type macro-F1	Any-event F1	High-intensity F1	Event-type macro-F1
Linear	0.7451	0.6371	0.5648	0.8179	0.8548	0.6887
LightGBM	0.8036	0.7948	0.6791	0.8197	0.8407	0.7008
CatBoost	0.8092	0.8087	0.6894	0.8229	0.8493	0.7201
GRU	0.8079	0.8245	0.6774	0.8162	0.8523	0.6955
TCN	0.8079	0.8086	0.6578	0.8203	0.8270	0.6763

Findings



Table 3: Lagged-ACLED-enhanced CatBoost under full modality availability, reported separately for week+1 through week+5.

Forecast horizon	Any-event F1	High-intensity F1	Event-type macro-F1
week+1	0.8256	0.8643	0.7252
week+2	0.8172	0.8468	0.7224
week+3	0.8276	0.8581	0.7227
week+4	0.8239	0.8536	0.7133
week+5	0.8200	0.8240	0.7167

Table 4: Missing-modality ablation for the base multimodal feature set, averaged over all five model families and all five forecast horizons. Positive deltas indicate higher F1 than the full-input condition.

Input configuration	Any-event		High-intensity		Event-type macro	
	F1	Δ vs full	F1	Δ vs full	F1	Δ vs full
Full input	0.7947	0.0000	0.7748	0.0000	0.6537	0.0000
No Black Marble NTL	0.7941	-0.0007	0.7788	+0.0040	0.6598	+0.0061
No NASA FIRMS	0.7996	+0.0049	0.7643	-0.0104	0.6619	+0.0082
No NTL + FIRMS	0.8009	+0.0061	0.7677	-0.0070	0.6859	+0.0322
No UNOSAT	0.7915	-0.0033	0.7536	-0.0212	0.6500	-0.0037

Limitations and Future Work

Current limitations

- The system forecasts access-related disruption signals, not direct humanitarian access ground truth
- Weekly aggregation and raion-level modeling may smooth short-lived local changes
- Satellite features are indirect proxies
- UNOSAT inputs are static pre-2024
- High-intensity is a rule-based target

Future work

- Add more direct access-ground-truth signals
- Incorporate more frequently updated damage and infrastructure layers
- Improve uncertainty calibration
- Explore finer spatiotemporal forecasting beyond the current raion-week setting

AccessWatch demonstrates
that a rapidly developed
research system can still
deliver rigorous evaluation,
meaningful empirical insight,
and a credible path toward
more operational
humanitarian early warning
tools.

Demo



THANK YOU!

Any Questions?